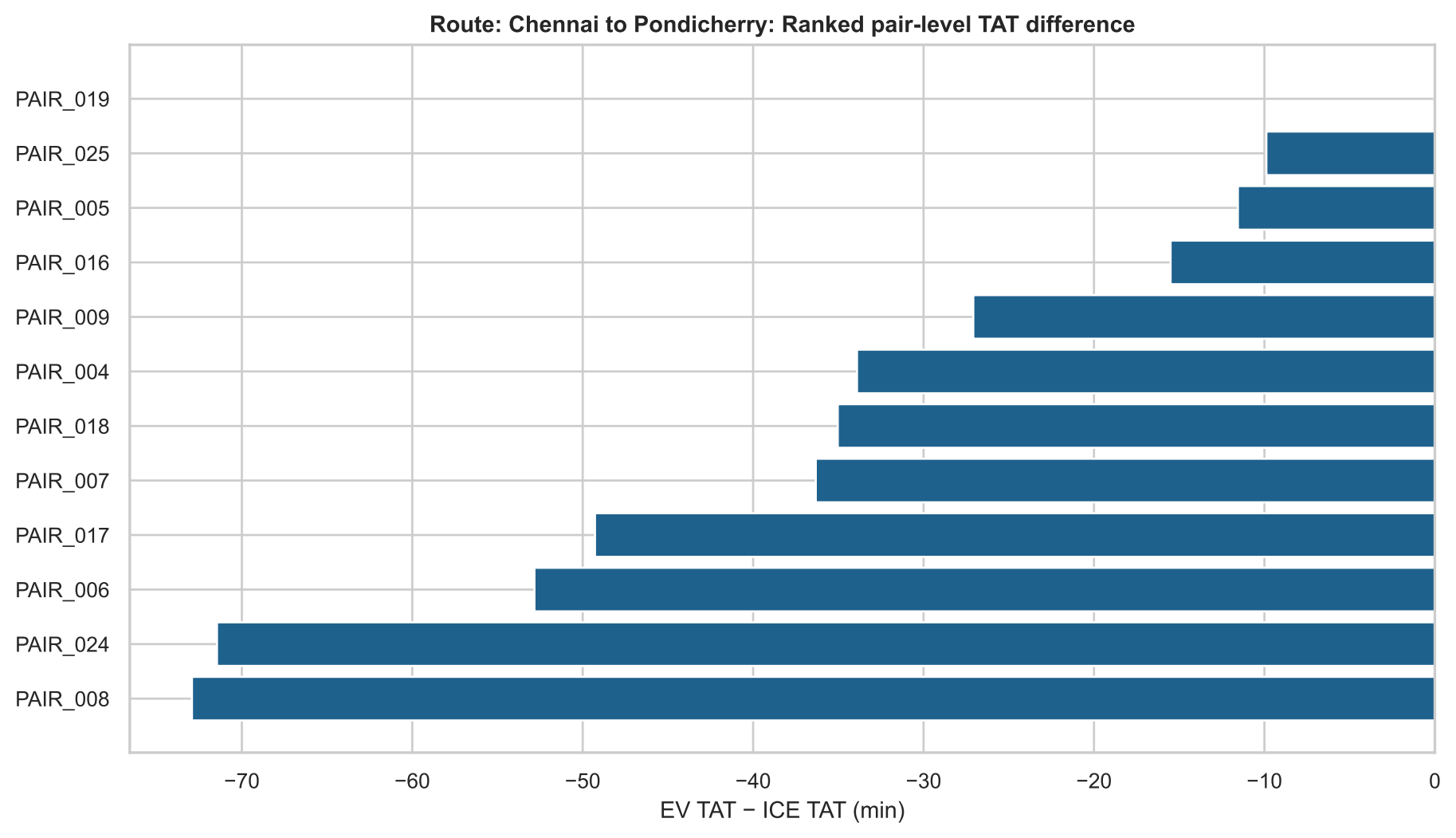
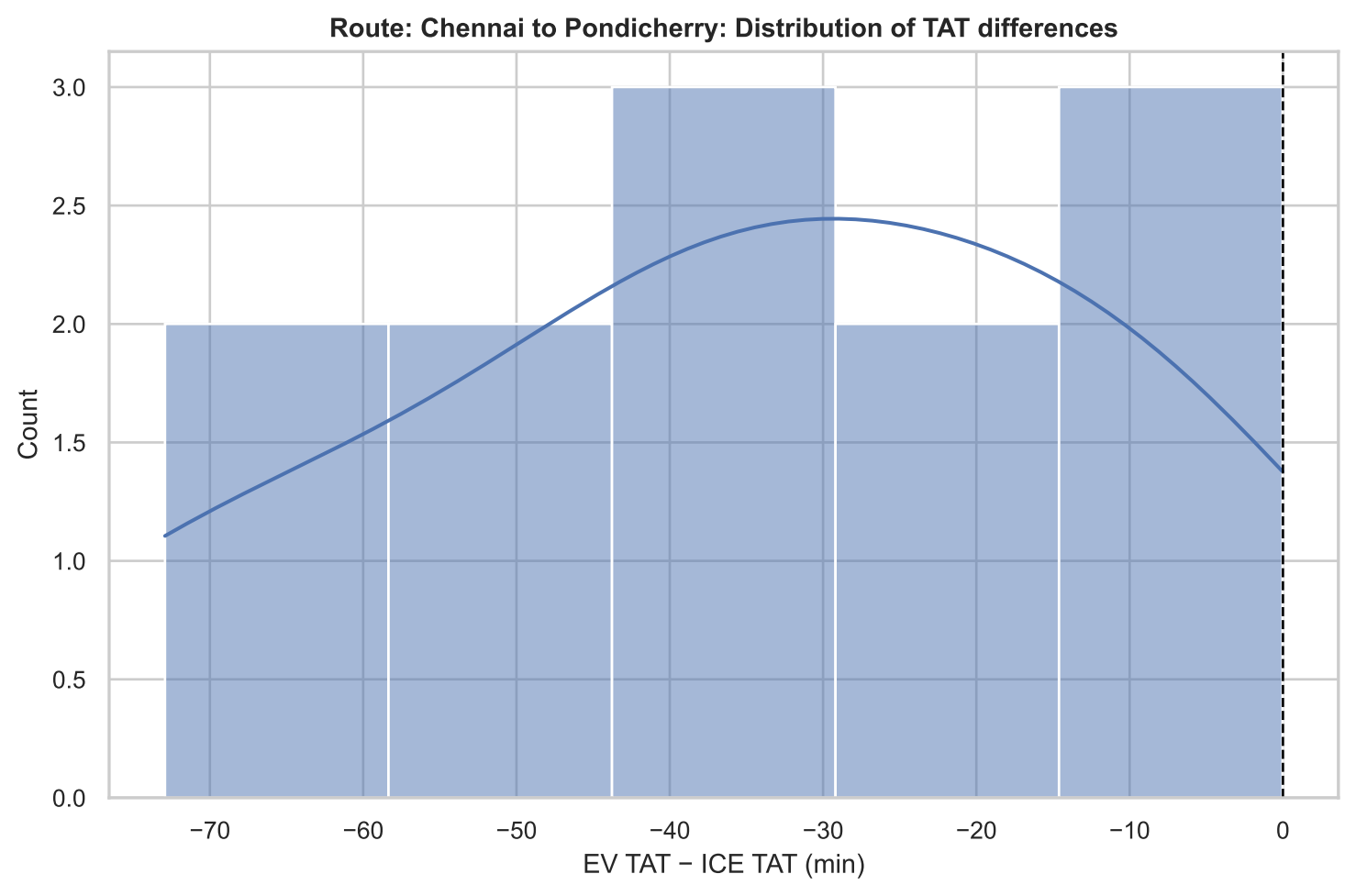
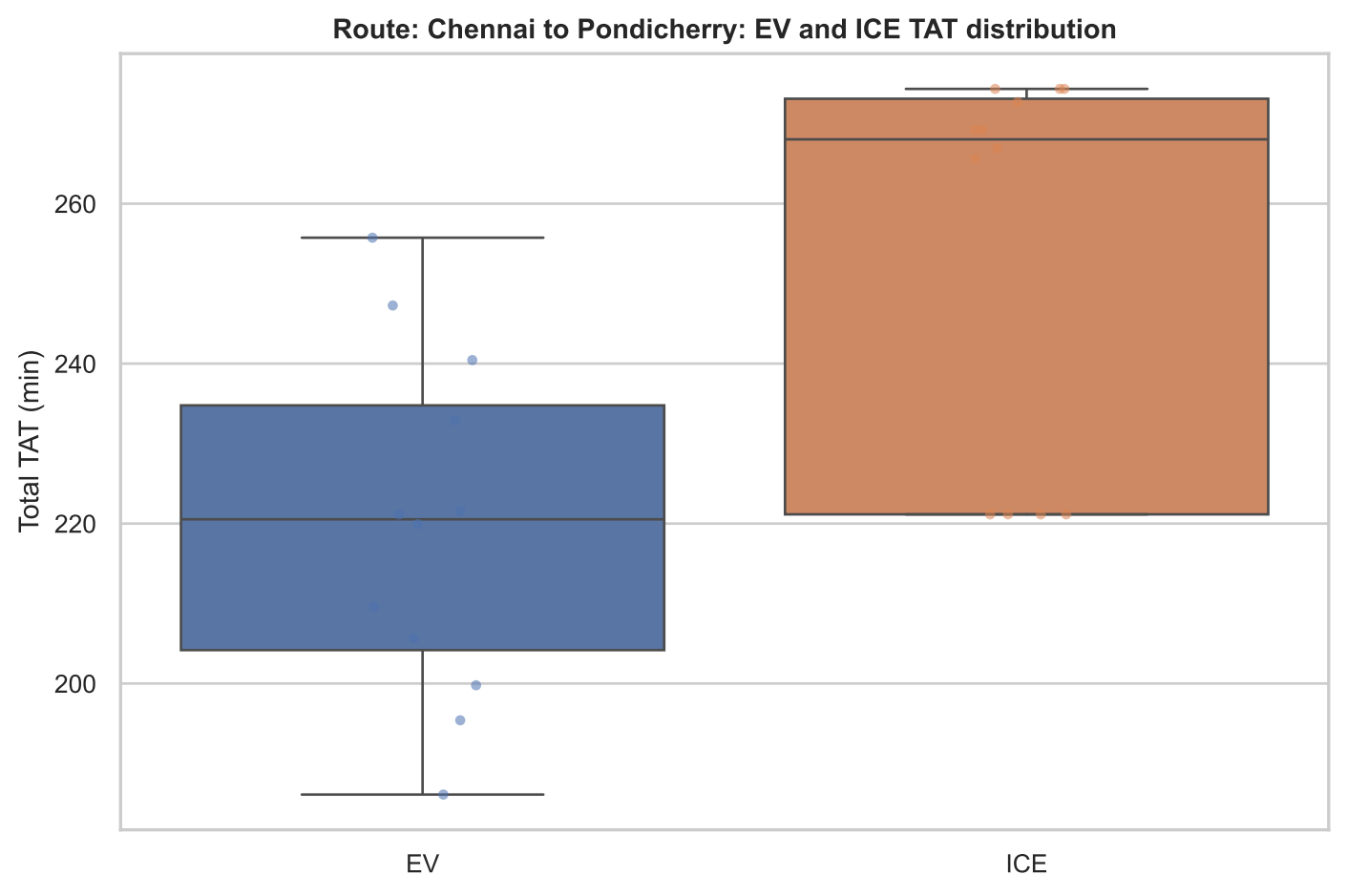
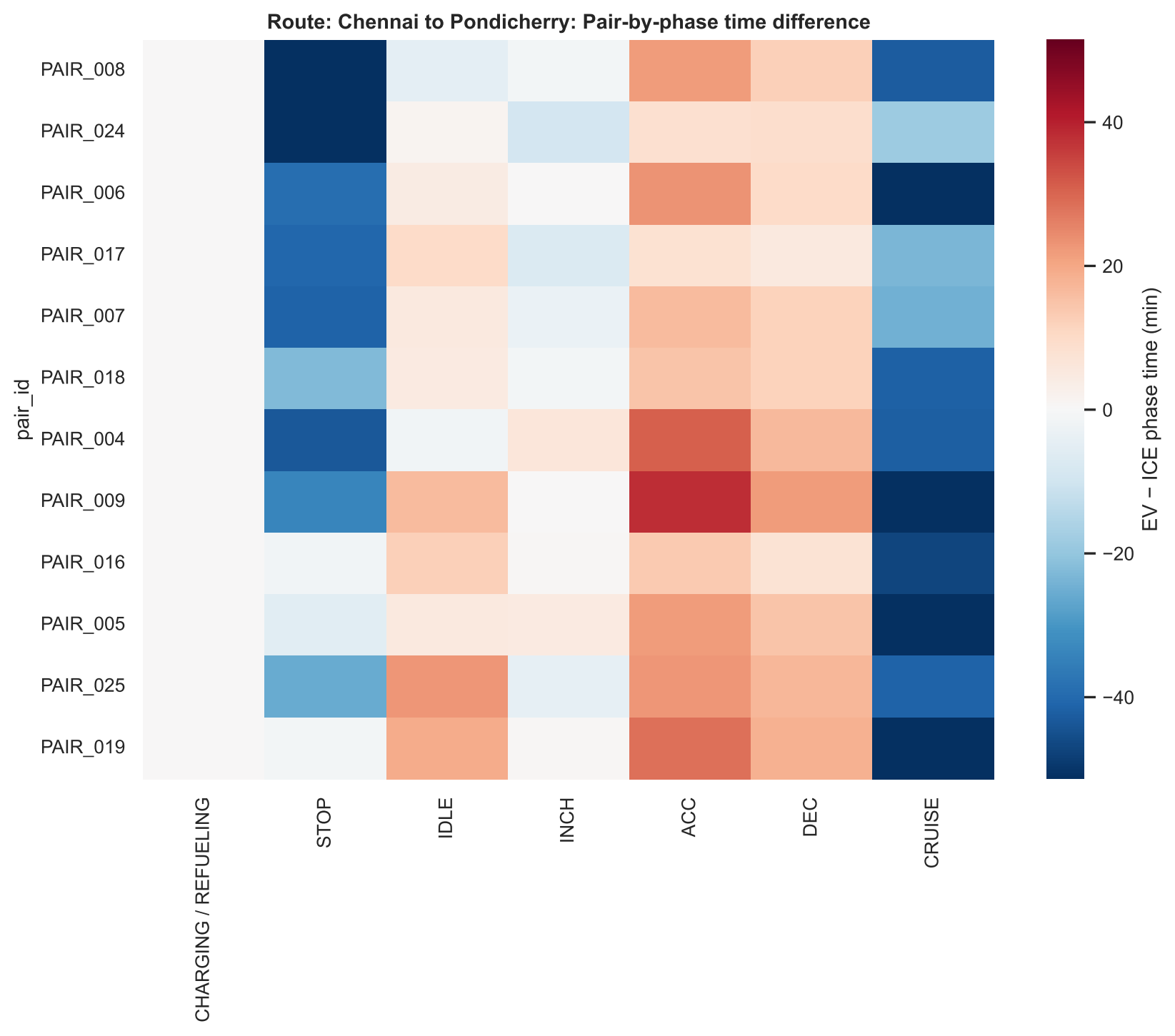
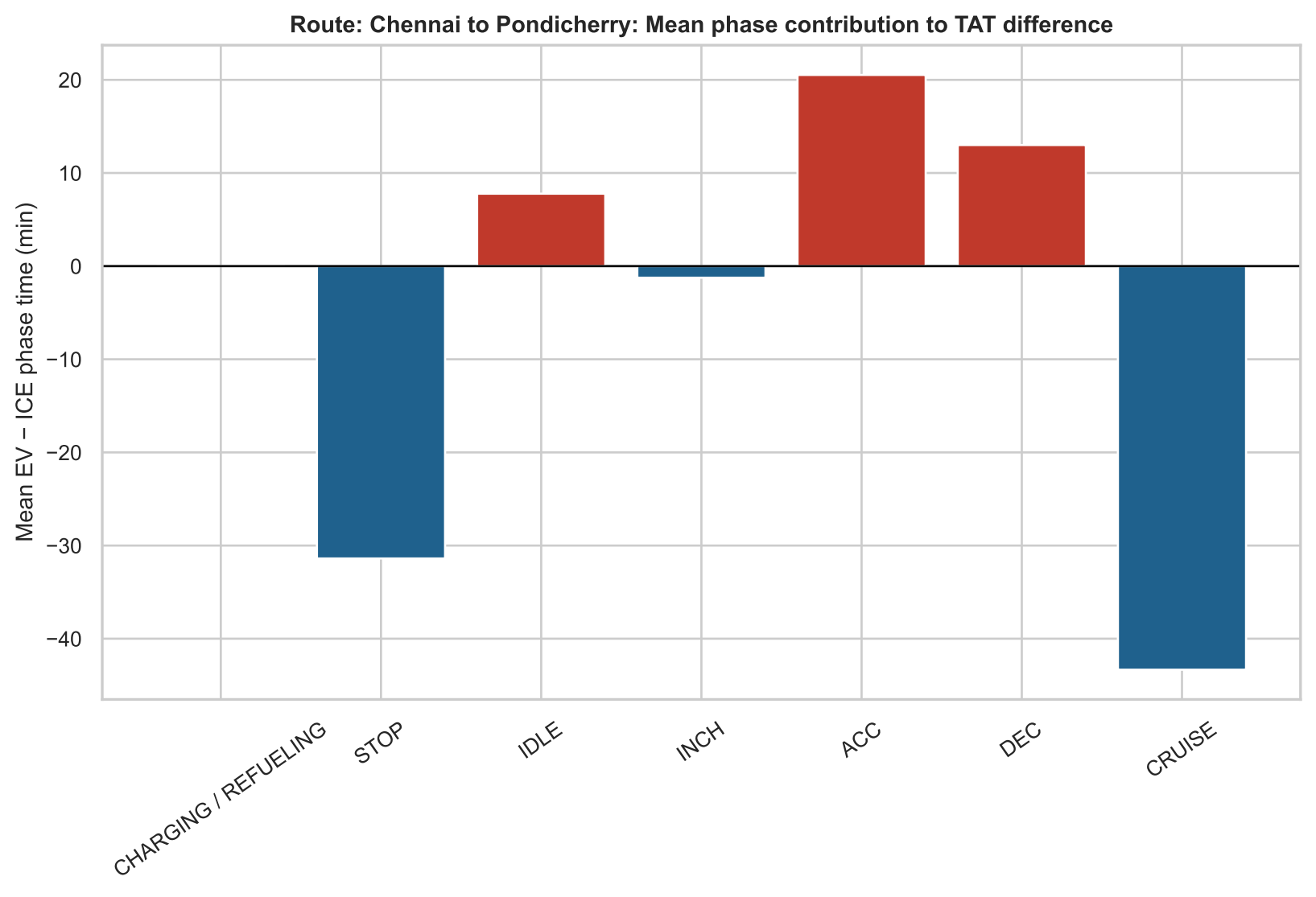
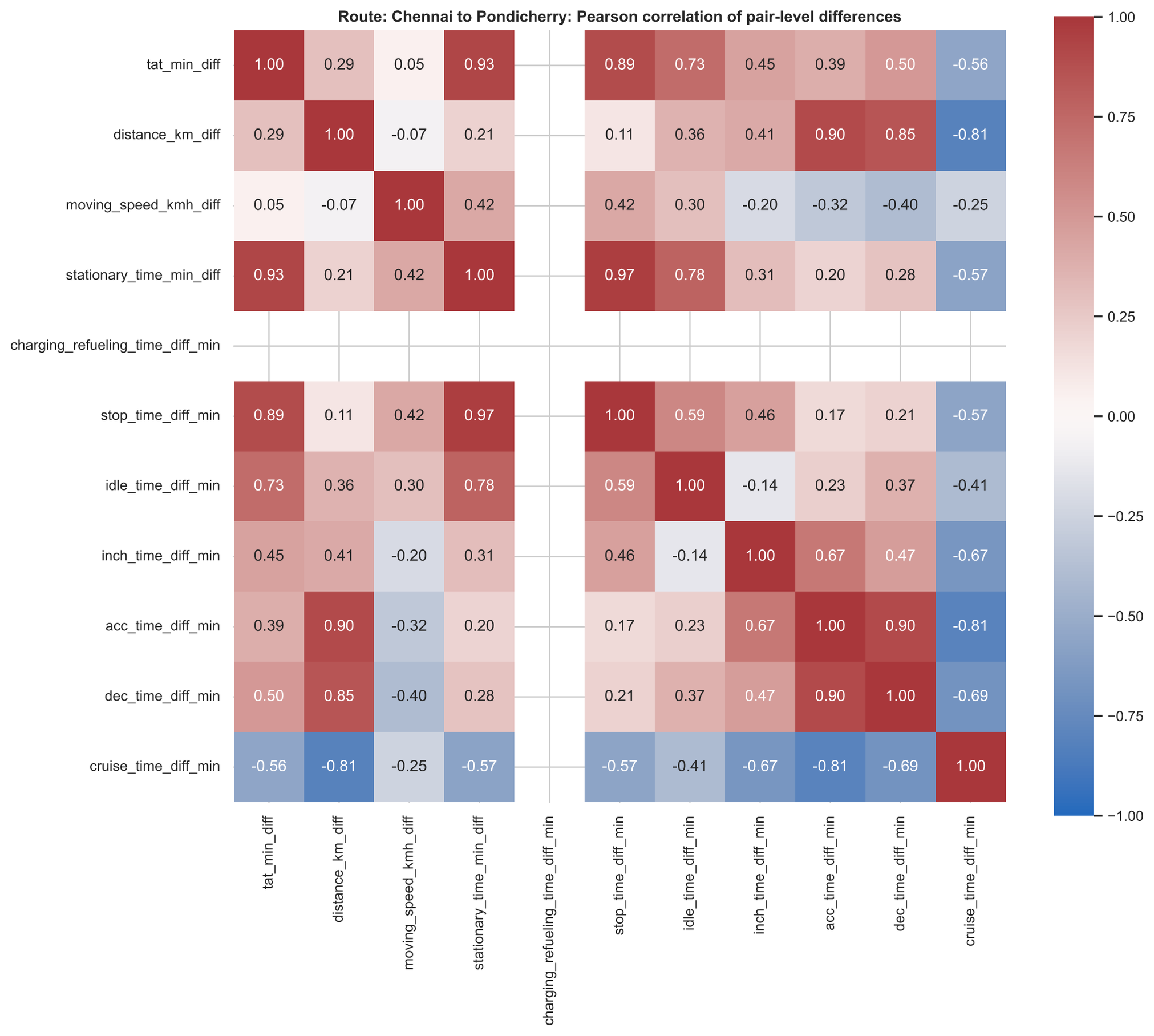

Route: Chennai to Pondicherry: Ranked pair-level TAT difference
PAIR_019
PAIR_025
PAIR_005
PAIR_016
PAIR_009
PAIR_004
PAIR_018
PAIR_007
PAIR_017
PAIR_006
PAIR_024
PAIR_008
−70
−60
−50
−40
−30
−20
−10
0
EV TAT − ICE TAT (min)

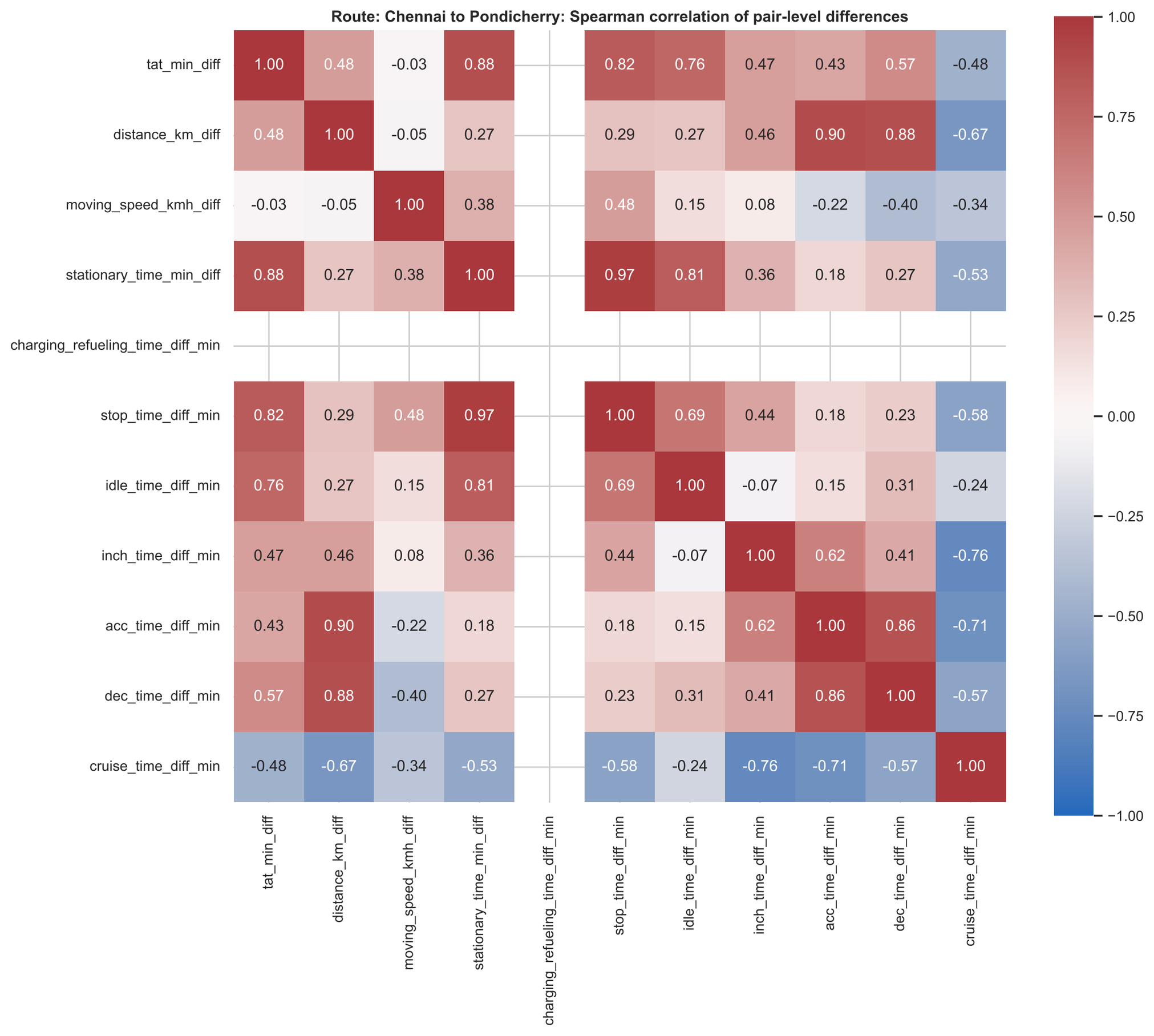

Route: Chennai to Pondicherry: Distribution of TAT differences
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−70
−60
−50
−40
−30
−20
−10
0
EV TAT − ICE TAT (min)

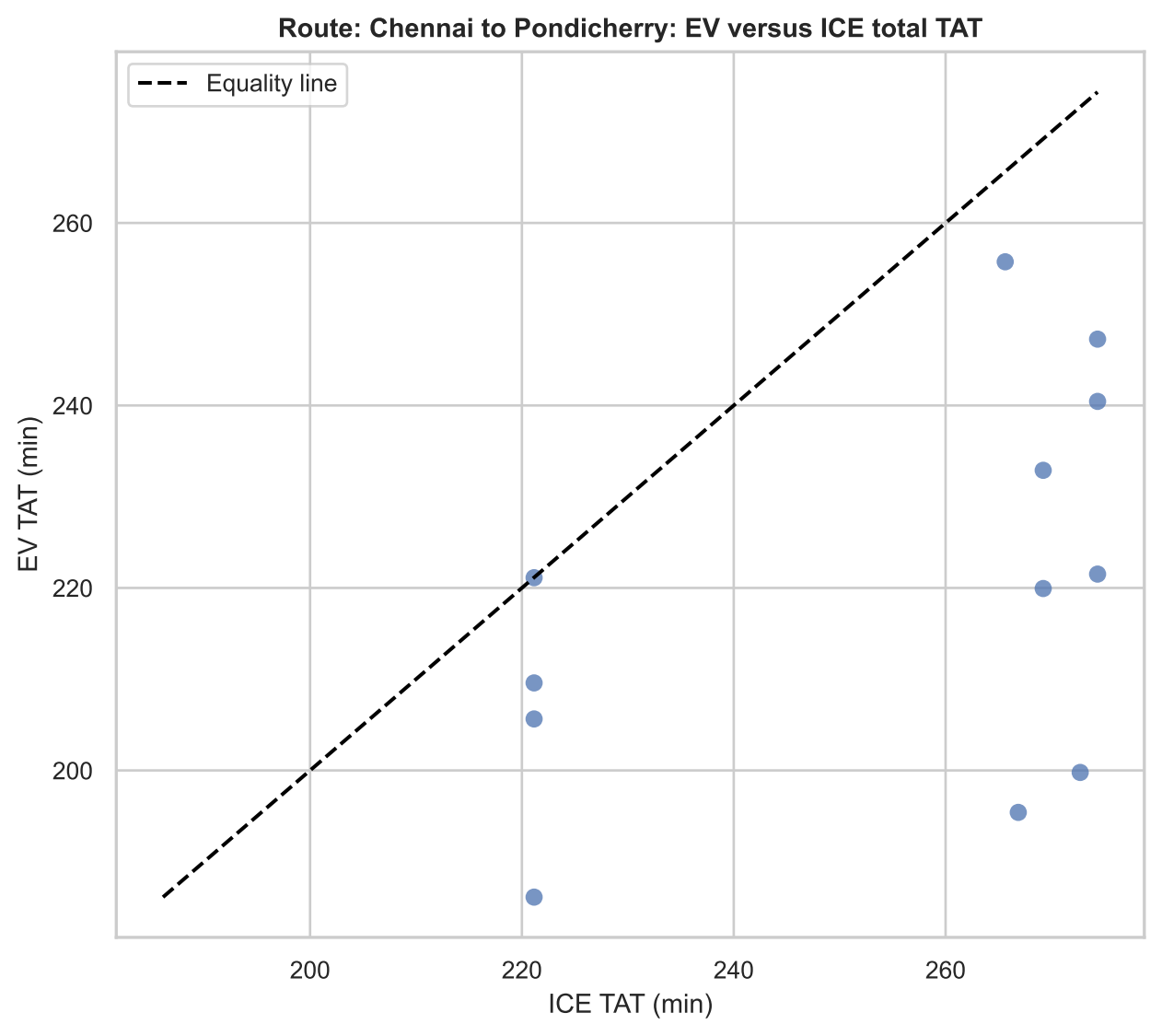

Route: Chennai to Pondicherry: EV and ICE TAT distribution
Total TAT (min)
260
240
220
200
EV
ICE

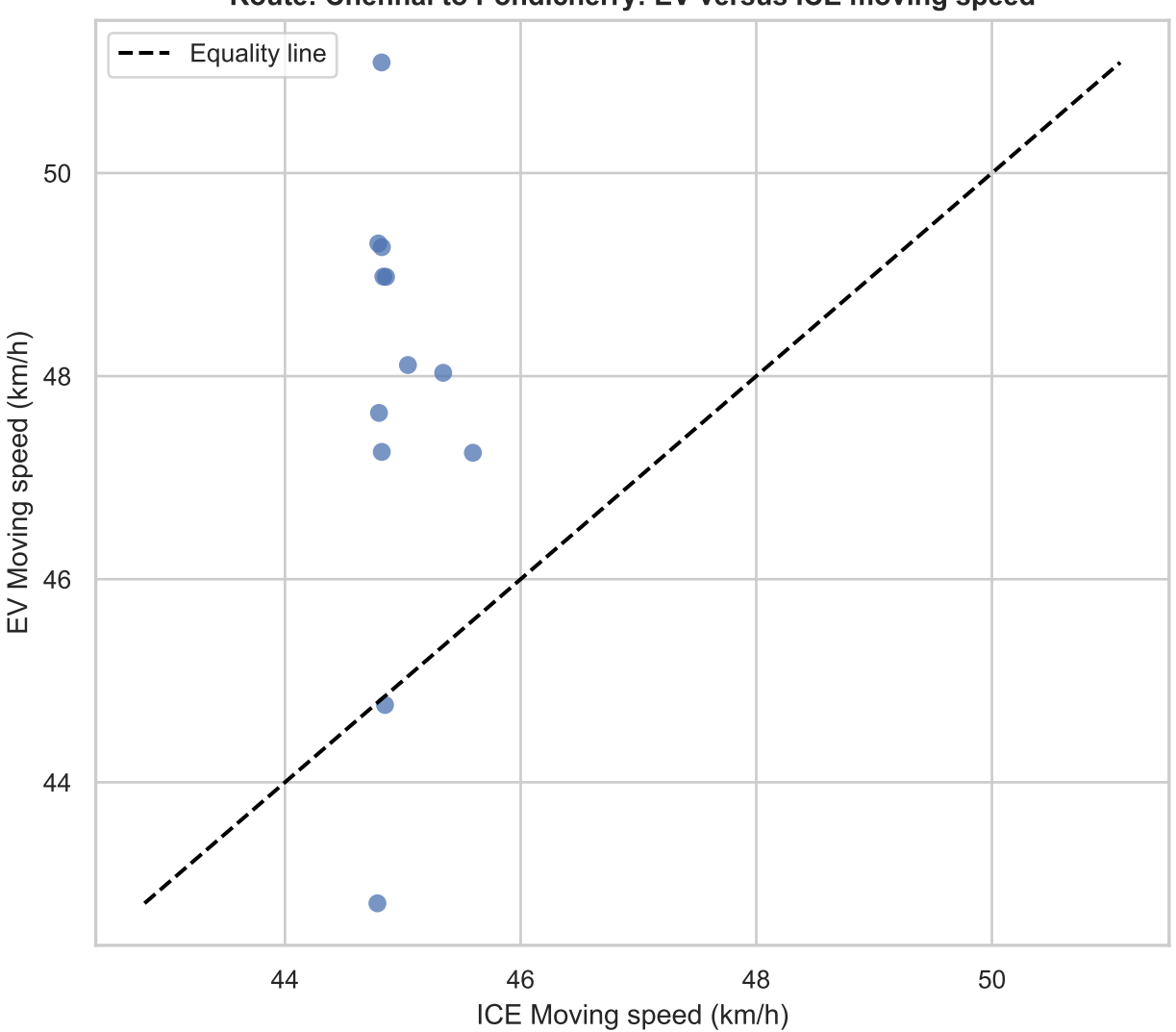

Route: Chennai to Pondicherry: Pair-by-phase time difference
pair_id
PAIR_008
PAIR_024
PAIR_006
PAIR_017
PAIR_007
PAIR_018
PAIR_004
PAIR_009
PAIR_016
PAIR_005
PAIR_025
PAIR_019
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
40
20
0
−20
−40

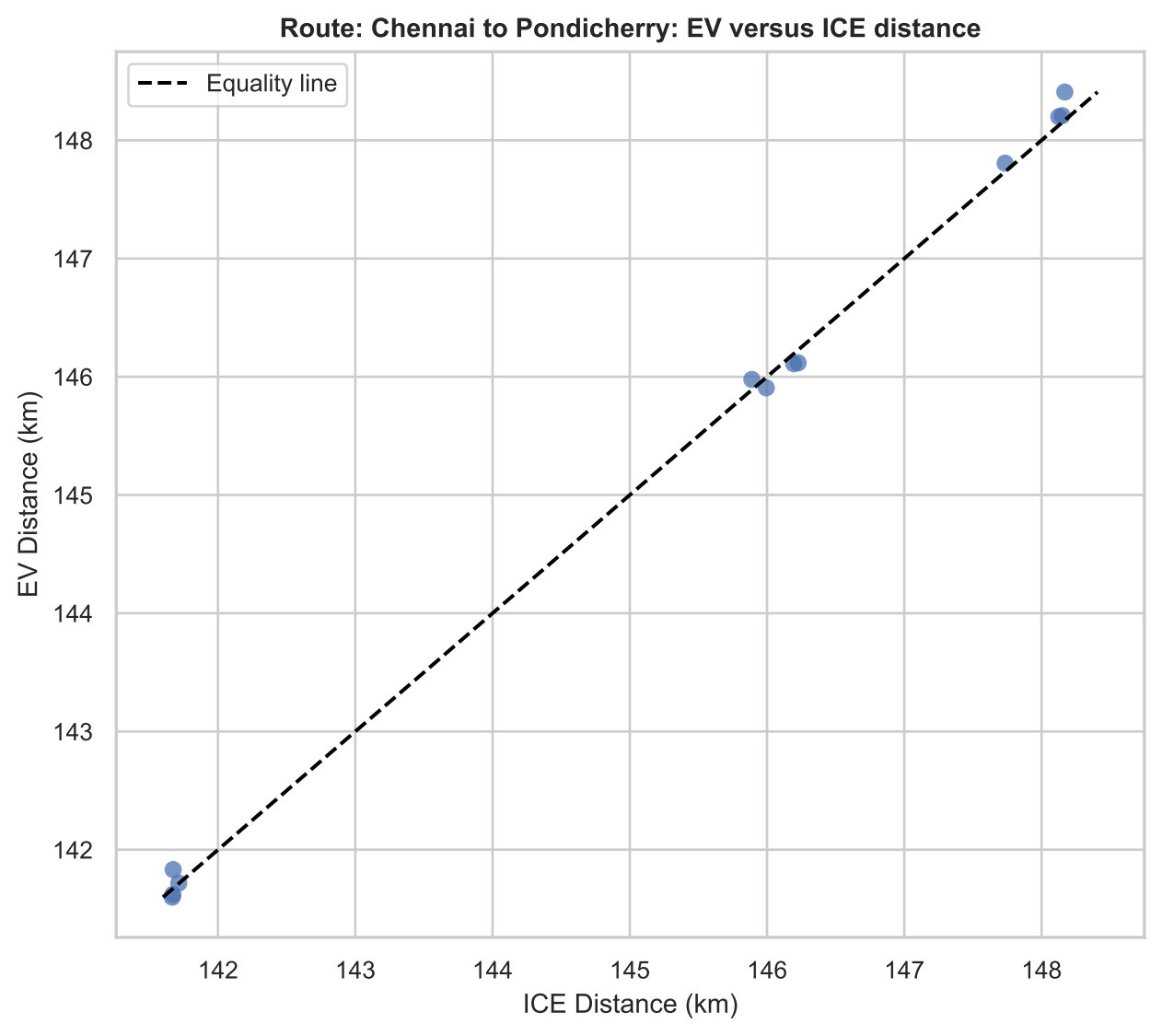

Route: Chennai to Pondicherry: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
20
10
0
−10
−20
−30
−40
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

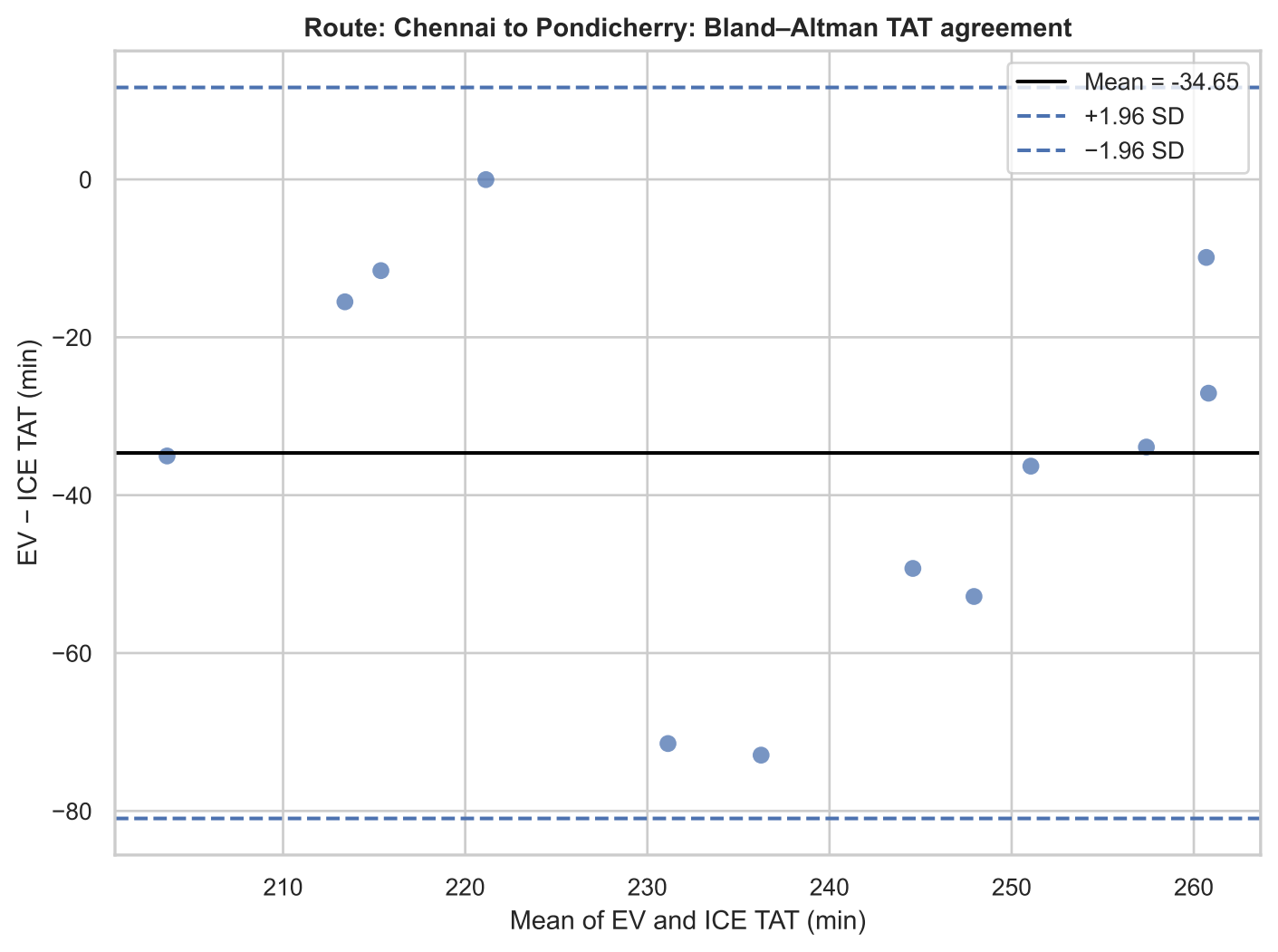

Route: Chennai to Pondicherry: Pearson correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 0.29 0.05 0.93 0.89 0.73 0.45 0.39 0.50 -0.56
0.29 1.00 -0.07 0.21 0.11 0.36 0.41 0.90 0.85 -0.81
0.05 -0.07 1.00 0.42 0.42 0.30 -0.20 -0.32 -0.40 -0.25
0.93 0.21 0.42 1.00 0.97 0.78 0.31 0.20 0.28 -0.57
0.89 0.11 0.42 0.97 1.00 0.59 0.46 0.17 0.21 -0.57
0.73 0.36 0.30 0.78 0.59 1.00 -0.14 0.23 0.37 -0.41
0.45 0.41 -0.20 0.31 0.46 -0.14 1.00 0.67 0.47 -0.67
0.39 0.90 -0.32 0.20 0.17 0.23 0.67 1.00 0.90 -0.81
0.50 0.85 -0.40 0.28 0.21 0.37 0.47 0.90 1.00 -0.69
-0.56 -0.81 -0.25 -0.57 -0.57 -0.41 -0.67 -0.81 -0.69 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

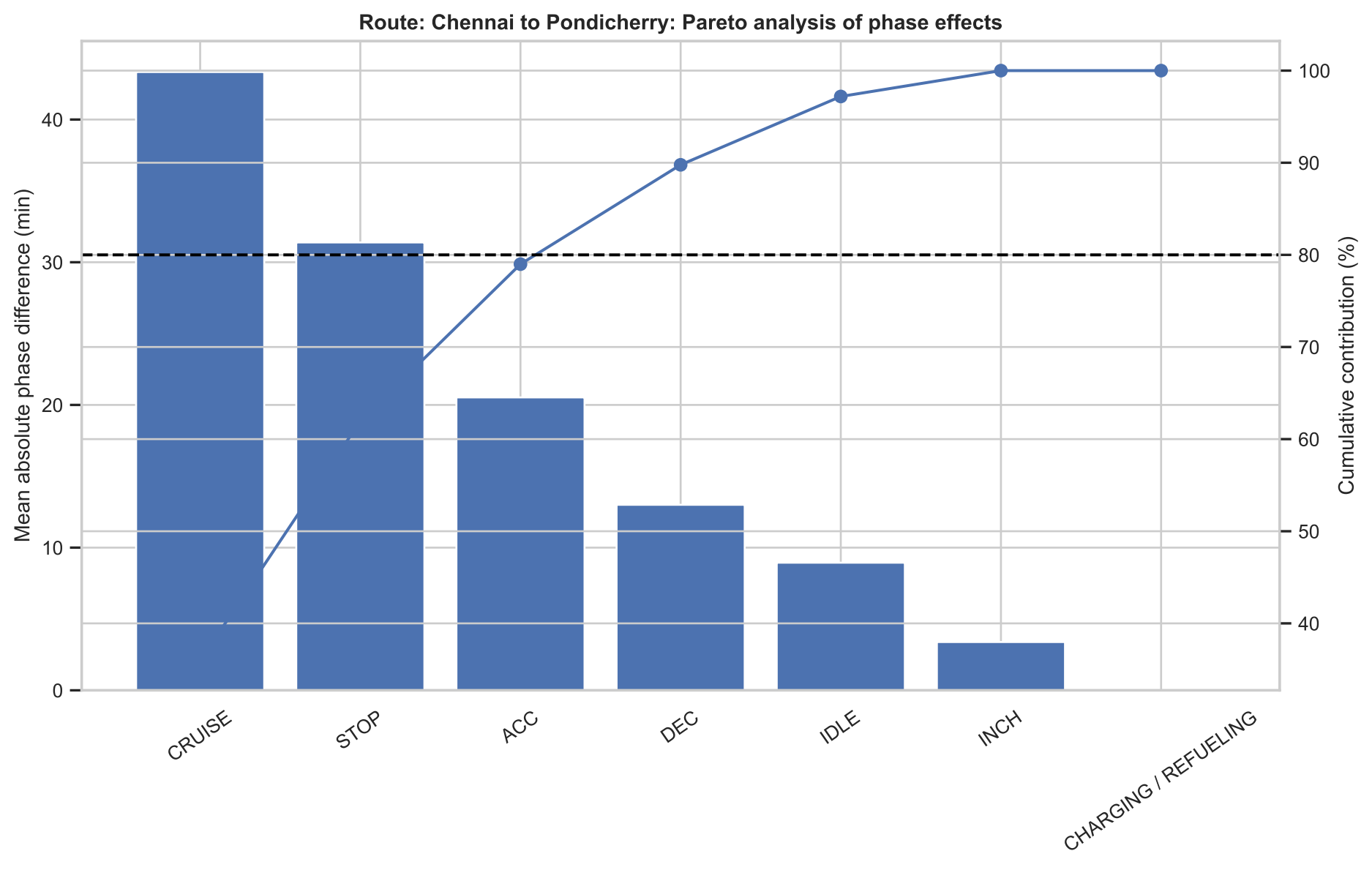

Route: Chennai to Pondicherry: Spearman correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 0.48 -0.03 0.88 0.82 0.76 0.47 0.43 0.57 -0.48
0.48 1.00 -0.05 0.27 0.29 0.27 0.46 0.90 0.88 -0.67
-0.03 -0.05 1.00 0.38 0.48 0.15 0.08 -0.22 -0.40 -0.34
0.88 0.27 0.38 1.00 0.97 0.81 0.36 0.18 0.27 -0.53
0.82 0.29 0.48 0.97 1.00 0.69 0.44 0.18 0.23 -0.58
0.76 0.27 0.15 0.81 0.69 1.00 -0.07 0.15 0.31 -0.24
0.47 0.46 0.08 0.36 0.44 -0.07 1.00 0.62 0.41 -0.76
0.43 0.90 -0.22 0.18 0.18 0.15 0.62 1.00 0.86 -0.71
0.57 0.88 -0.40 0.27 0.23 0.31 0.41 0.86 1.00 -0.57
-0.48 -0.67 -0.34 -0.53 -0.58 -0.24 -0.76 -0.71 -0.57 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

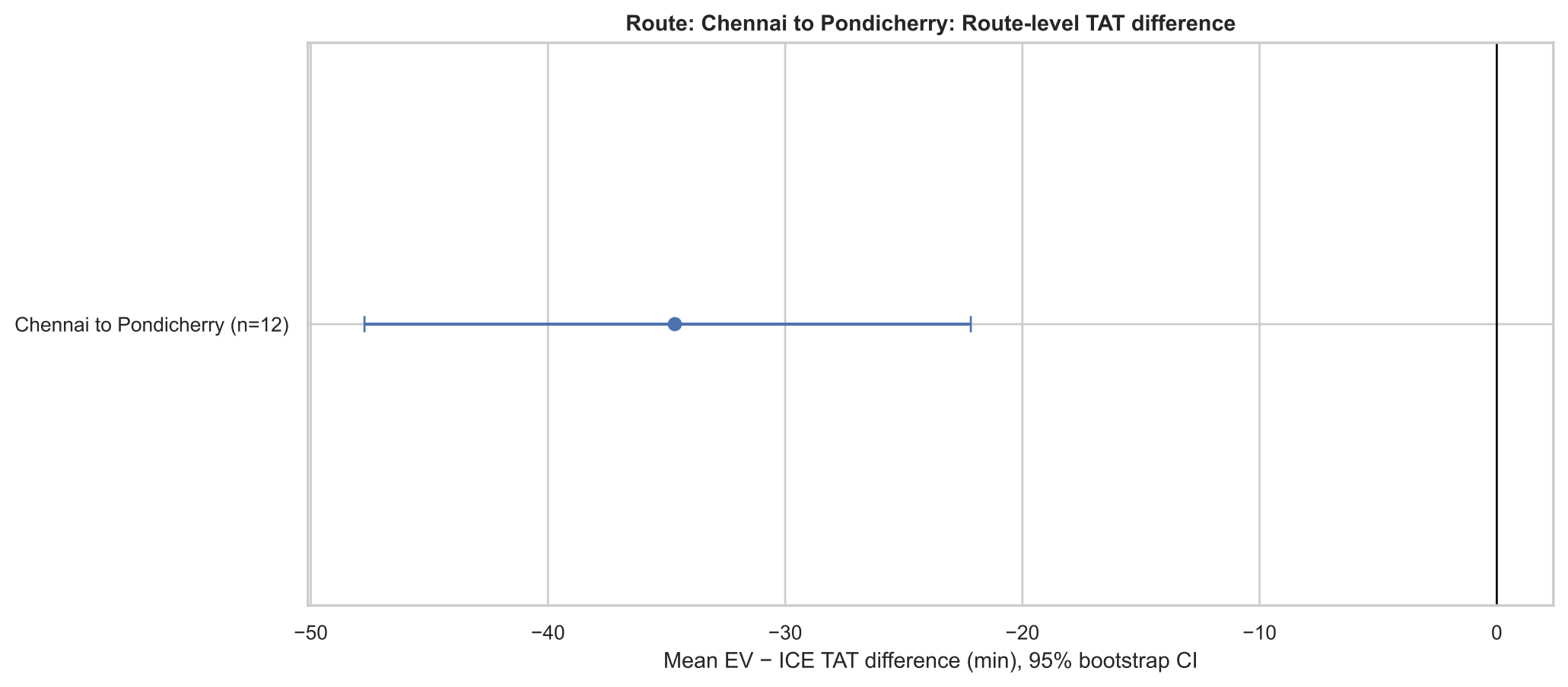

Route: Chennai to Pondicherry: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
200
220
240
260
200
220
240
260

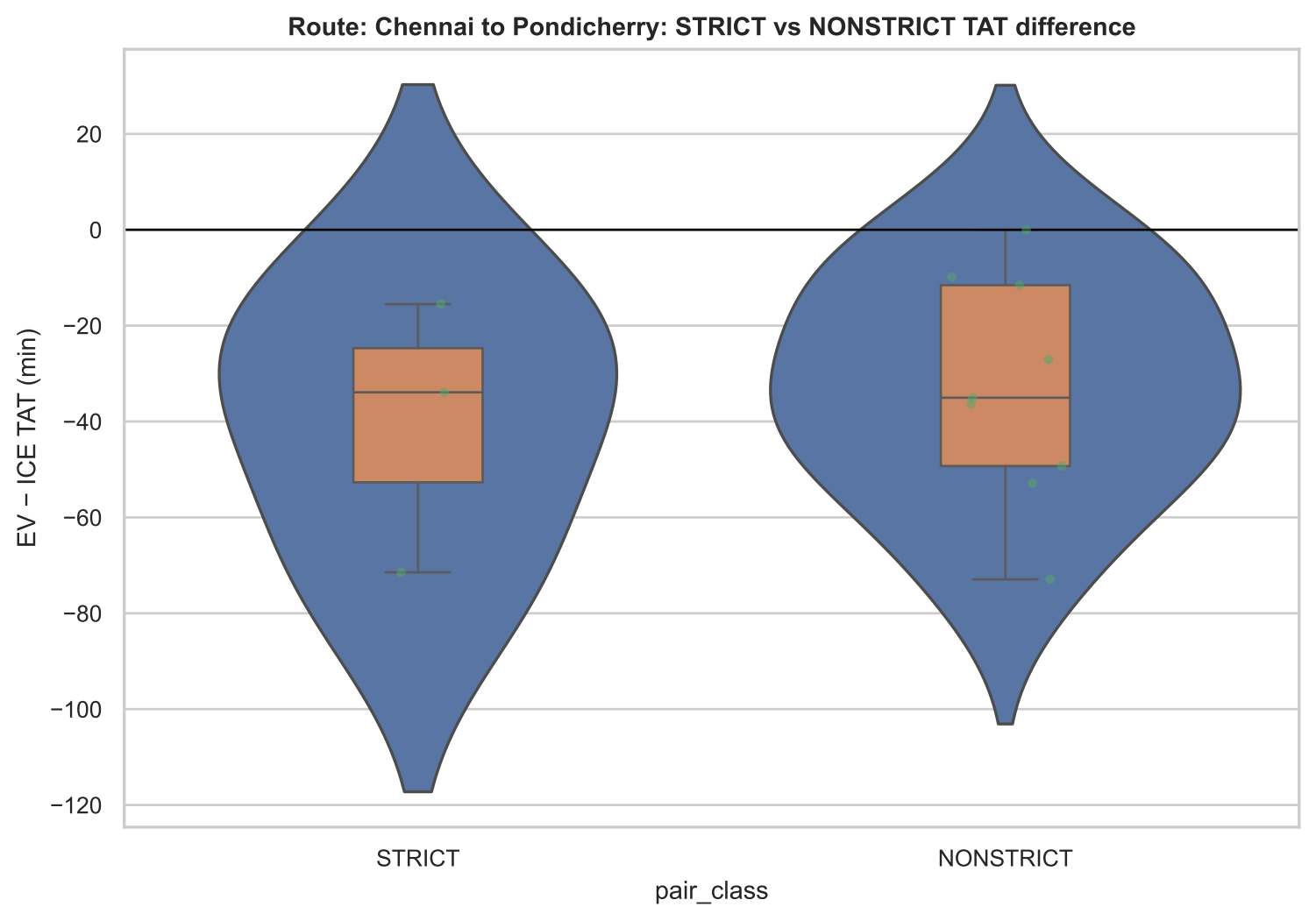

Route: Chennai to Pondicherry: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
44
46
48
50
44
46
48
50

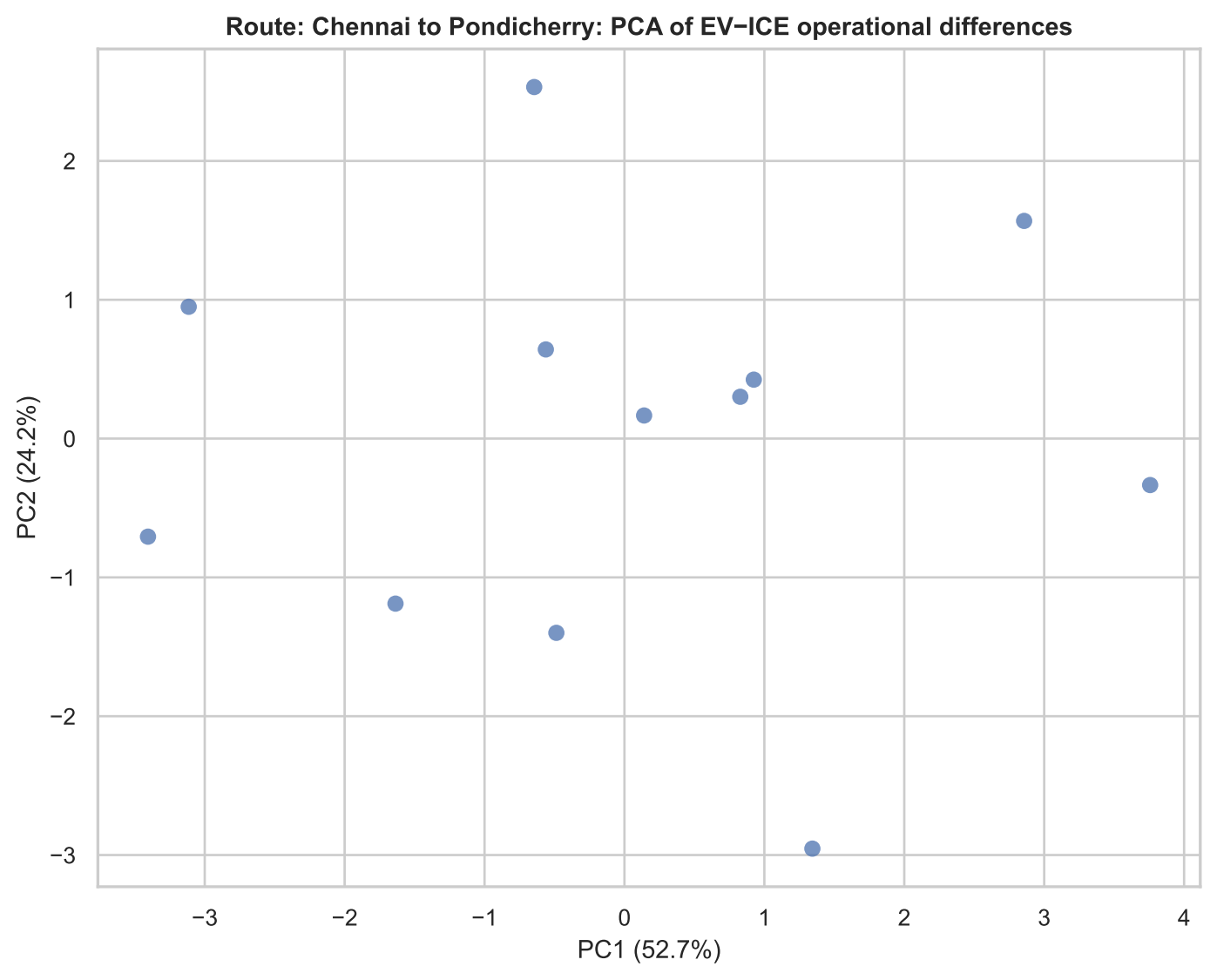

Route: Chennai to Pondicherry: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
142
143
144
145
146
147
148

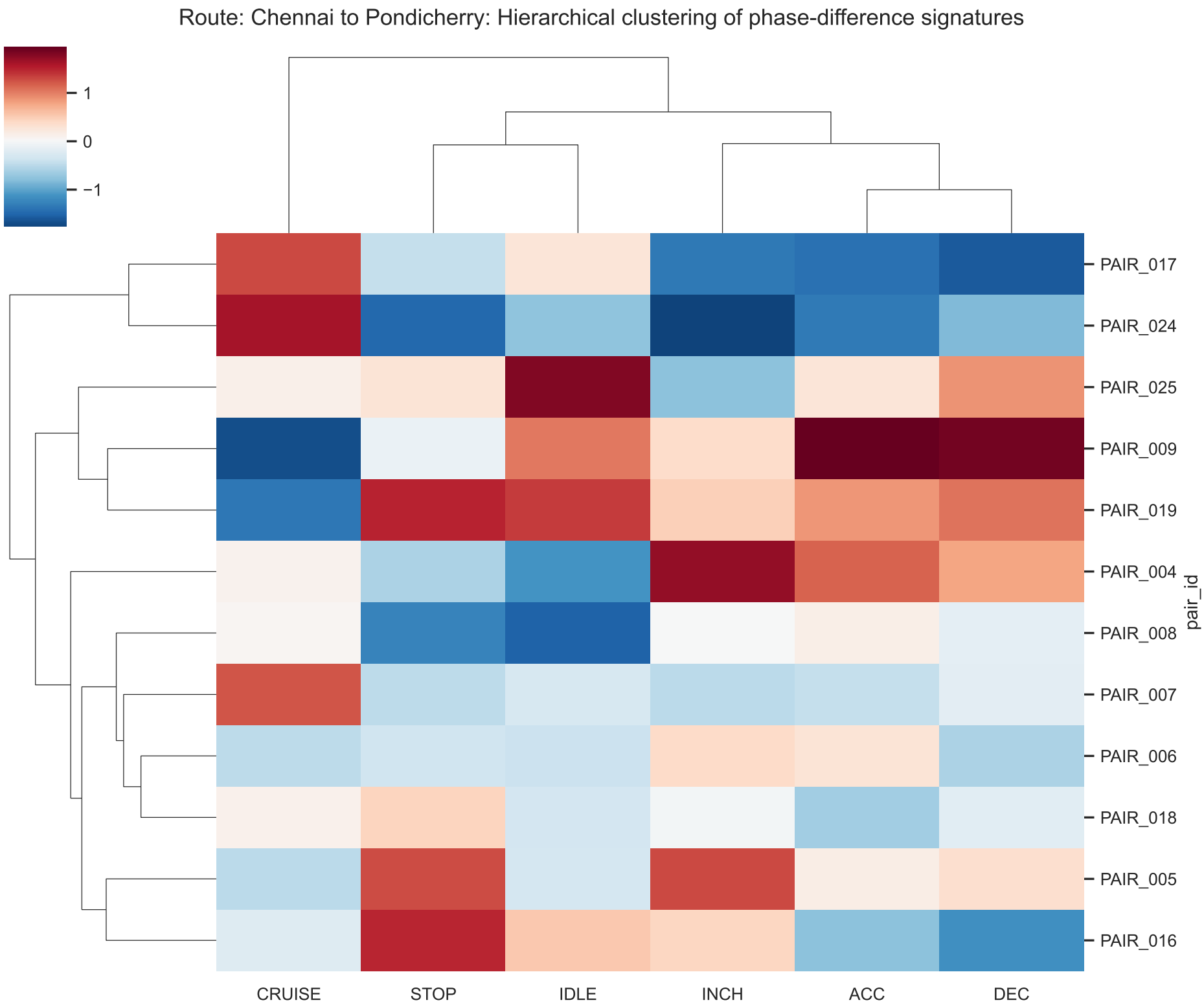

Route: Chennai to Pondicherry: Bland–Altman TAT agreement
Mean = -34.65
+1.96 SD
−1.96 SD
0
−20
−40
−60
−80
EV – ICE TAT (min)
210
220
230
240
250
260
Mean of EV and ICE TAT (min)


Route: Chennai to Pondicherry: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
10
20
30
40
40
50
60
70
80
90
100
CRUISE
STOP
ACC
DEC
IDLE
INCH
CHARGING / REFUELING


Route: Chennai to Pondicherry: Route-level TAT difference
Chennai to Pondicherry (n=12)
−50
−40
−30
−20
−10
0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Chennai to Pondicherry: STRICT vs NONSTRICT TAT difference
20
0
−20
−40
−60
−80
−100
−120
EV – ICE TAT (min)
STRICT
NONSTRICT
pair_class


Route: Chennai to Pondicherry: PCA of EV−ICE operational differences
PC2 (24.2%)
PC1 (52.7%)
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3
4


Route: Chennai to Pondicherry: Hierarchical clustering of phase-difference signatures
1
0
−1
PAIR_017
PAIR_024
PAIR_025
PAIR_009
PAIR_019
PAIR_004
PAIR_008
PAIR_007
PAIR_006
PAIR_018
PAIR_005
PAIR_016
pair_id
CRUISE
STOP
IDLE
INCH
ACC
DEC